

Diesel Engines**V 4000 R 10****V 4000 R 20****Maintenance Schedule****M050478/01E**

Low operating and maintenance costs as well as operational reliability and availability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.

The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble-free engine operation at all times. For this purpose always:

- ensure that sufficient fuel is available,
- ensure that the combustion air is dry and clean.

Our Product Support Service will always be available should assistance be required.

Preventive maintenance instructions.

- Special care should be taken to keep the machinery plant in a clean and serviceable condition at all times to facilitate early detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic components from oil and fuel, never treat with organic detergents. Wipe with a dry cloth only.
- Always replace all seals and gaskets.

MTU Maintenance Concept

The maintenance concept for MTU products is a preventive maintenance concept.

Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability.

The Maintenance Schedule is based on the power output defined in the individual orders, under consideration of the operating parameters established in the order's specifications.

The time intervals according to which the maintenance tasks and the relevant checks and tasks involved are to be completed are based on operational experience and can only serve as guidelines. Specific operational conditions may require modification of the Maintenance Schedule.

Note:

The engine manufacturer cannot be held responsible for the performance of the following maintenance tasks.

The operator is responsible for carrying out the following maintenance tasks:

Fuel filter

The maintenance interval depends on the degree of fuel contamination. The fuel filter must be replaced after 2 years at the latest.

Battery

Battery maintenance depends on the degree of use and ambient temperature. The battery has to be checked according to the manufacturer's instructions, as it is not always a part of the MTU scope of supply.

Fluids and Lubricants:

Only those fluids and lubricants are to be used that are approved in the MTU Fluids and Lubricants Specification A001061. A special note must be made that, after maintenance or repair work on the cooling system, the operator must check the concentration of the corrosion inhibitor or the corrosion-inhibitor antifreeze.

Major Overhaul

Requires complete engine disassembly to facilitate repair/overhaul of all relevant components.

Out-of-Service Periods

If the engine is to remain out of service for several weeks, spray engine oil into the air-intake ducting. Actuate the engine shutdown system and bar the engine with the starting equipment when performing this task. Carry out maintenance tasks in accordance with the time limit schedule (see the Maintenance Frequency Chart). In addition, the engine must be barred manually every month. If the engine is to remain out of service for more than 1 month, carry out engine preservation in accordance MTU Fluids and Lubricants Specification, Publication No. A001061.

Application group

2A: Rail Traction

Maintenance Frequency Chart

Maintenance Level	W1: Operational checks, daily	X
	W2: Operating hours	500
	Limit, months	6
	W3: Operating hours	1 000
	Limit, year	1
	W4: Operating hours	5 000
	Limit, year(s)	2
	W5: Operating hours	10 000
	Limit, year(s)	6
	W6: Operating hours	30 000
	Limit, year(s)	18

Maintenance Task Schedule

Code No.	Maintenance Level W1 – Operational checks	
G000.000.13	Engine operation	Check speed
G000.000.15	Engine operation	Check pressures (if gauges are provided)
G000.000.17	Engine operation	Check temperatures (if gauges are provided)
G000.000.19	Engine operation	Check running noises
G000.000.21	Engine operation	Check engine and external pipework for leaks
G101.011.05	Turbocharger connections	Check exhaust connections for leaks Check oil supply/return pipework for leaks
G121.052.01	Air filter	Check service indicator
G124.051.01	Intake air line	Check drain line for water discharge and obstructions
G140.000.05	Exhaust system	Check exhaust gas color
G140.000.11	Exhaust system	Drain condensate (if drain valve provided)
G180.000.01	Engine oil	Check level
G202.000.03	Engine coolant	Check level
G510.070.01	Monitoring system	Carry out lamp test